

PYEONGTAEK AB, Korea, Republic ofWMO: **471270**Lat: **36.967N**Lon: **127.033E**Elev: **52**StdP: **14.67**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **43216****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.2	13.7	-0.8	5.3	17.2	3.1	6.4	19.8	22.0	27.1	19.2	26.7	4.1	320	0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.4	91.6	78.9	89.4	77.6	87.0	76.5	81.7	86.8	80.1	85.5	78.9	84.0	7.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.7	159.7	83.8	78.9	150.4	82.5	77.3	142.2	81.3	45.7	87.3	44.0	86.1	42.5	82.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
18.4	15.8	13.7	DB	4.3	95.1	4.8	2.5	0.8	96.9	-2.0	98.3	-4.7	99.8	-8.2	101.6	
			WB	3.2	84.6	4.6	2.3	-0.2	86.2	-2.9	87.5	-5.5	88.8	-8.9	90.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	54.9	27.7	32.3	42.1	54.0	64.5	72.7	78.6	79.6	70.5	58.4	44.8	31.8	(6)
		DBStd	18.97	7.24	6.98	6.44	6.34	4.75	3.62	3.98	4.43	5.01	6.35	7.68	7.52	
		HDD50	2240	691	495	256	31	0	0	0	0	12	192	563		
		HDD65	5046	1156	914	709	334	67	2	0	12	217	607	1028		
		CDD50	4022	0	1	12	151	451	681	888	917	614	272	35	0	
		CDD65	1349	0	0	0	3	52	232	423	452	176	11	0	0	
		CDH74	10599	0	0	1	53	419	1418	3510	4119	1016	62	1	0	
		CDH80	3357	0	0	0	5	62	350	1188	1538	212	2	0	0	
(14)	Wind	WSAvg	5.6	5.5	6.0	6.8	6.7	6.1	5.4	5.5	5.3	4.8	4.6	5.3	5.3	(14)
(15)	Precipitation	PrecAvg	50.2	0.7	1.1	1.8	3.1	3.4	5.6	12.6	10.8	6.4	2.0	1.7	1.1	(15)
(16)		PrecMax	74.1	2.2	3.3	4.6	7.2	8.4	15.1	27.4	38.9	14.3	8.0	5.5	2.1	(16)
(17)		PrecMin	29.8	0.1	0.0	0.4	0.9	0.5	0.9	4.8	1.8	0.6	0.0	0.2	0.3	(17)
(18)		PrecStd	10.6	0.6	0.8	1.0	1.5	1.7	3.6	4.9	7.8	4.6	1.8	1.2	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	50.4	58.5	68.2	79.3	85.3	89.4	95.0	95.4	87.9	79.0	69.7	55.6		(19)
(20)		0.4% MCWB	44.4	51.8	54.0	61.0	65.2	70.4	80.3	79.8	74.6	66.0	60.4	48.5		(20)
(21)		2% DB	45.0	51.9	62.7	74.8	81.2	86.2	91.4	92.9	84.6	75.4	64.6	50.2		(21)
(22)		2% MCWB	40.0	44.1	50.9	59.4	64.0	70.6	79.1	79.6	72.4	63.5	56.5	45.0		(22)
(23)		5% DB	41.9	47.9	58.6	71.0	78.7	84.1	89.0	90.3	82.4	72.4	61.0	46.8		(23)
(24)		5% MCWB	37.3	41.1	48.5	56.5	63.2	69.8	78.2	78.5	71.0	61.7	54.1	42.4		(24)
(25)		10% DB	39.0	44.2	54.2	66.6	75.6	82.0	86.3	87.9	80.2	69.8	57.5	44.2		(25)
(26)		10% MCWB	35.4	38.5	45.8	54.8	62.1	69.4	77.3	78.1	69.6	60.5	51.5	40.4		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	45.7	53.7	59.0	67.4	72.5	78.9	84.2	84.1	78.6	71.1	63.2	49.8		(27)
(28)		0.4% MCDB	48.2	56.9	63.7	73.0	77.2	81.4	88.8	89.0	85.1	75.0	66.0	52.8		(28)
(29)		2% WB	41.1	45.2	54.0	63.0	69.1	75.9	81.9	82.0	75.9	67.1	58.7	46.2		(29)
(30)		2% MCDB	44.0	49.7	59.1	69.2	75.2	79.9	86.4	87.5	81.1	71.5	62.7	49.0		(30)
(31)		5% WB	38.2	41.8	50.1	59.8	66.7	73.8	80.2	80.6	74.2	64.4	55.5	43.1		(31)
(32)		5% MCDB	40.7	46.4	55.7	66.6	73.2	78.5	85.1	86.5	78.6	69.4	59.5	46.1		(32)
(33)		10% WB	35.6	39.0	46.9	56.9	64.6	72.0	78.9	79.2	72.4	62.3	52.7	40.5		(33)
(34)		10% MCDB	38.1	42.6	52.4	63.4	71.4	77.4	83.6	85.0	76.9	67.2	56.6	43.4		(34)
(35)	Mean Daily Temperature Range	MDBR	17.4	18.8	20.2	22.2	20.4	17.4	13.8	14.4	17.3	20.4	18.8	17.2		(35)
(36)		5% DB MCDBR	19.5	23.8	26.8	28.5	25.0	21.3	17.4	17.2	19.2	22.4	22.1	20.4		(36)
(37)		5% DB MCWBR	14.9	16.8	16.3	14.8	11.3	9.2	8.1	7.5	9.4	12.0	15.0	15.8		(37)
(38)		5% WB MCDBR	17.4	20.9	22.5	22.5	19.5	16.2	14.4	14.9	15.6	18.4	19.2	19.1		(38)
(39)		5% WB MCWBR	14.7	16.1	15.9	14.4	11.7	9.4	8.3	7.9	9.5	11.9	14.8	15.9		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.442	0.490	0.572	0.607	0.622	0.693	0.652	0.579	0.507	0.494	0.478	0.439		(40)
(41)		taud	1.940	1.858	1.711	1.683	1.702	1.623	1.758	1.928	2.054	1.961	1.946	1.964		(41)
(42)		Ebn at Noon	220	227	220	223	223	207	214	227	237	223	207	211		(42)
(43)		Edh at Noon	45	55	70	76	75	82	71	59	49	49	44	41		(43)
(44)	All-Sky Solar Radiation	RadAvg	783	1031	1362	1584	1774	1640	1322	1391	1303	1141	783	689		(44)
(45)		RadStd	48	85	137	119	164	159	194	200	141	86	84	49		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	-2.25	N/A	N/A	N/A	N/A	N/A	+200	+119
(47)	Regional (49 neighbors)	N/A	N/A	-1.46	N/A	N/A	N/A	N/A	N/A	+166	+92

Nomenclature: See separate page